

The Exact Idempotent Gap Between Two Notions of Pure Infiniteness for Banach Algebras

Partial-result packet for arXiv:2406.05717

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Status: candidate substantial partial result, likely valid; human review requested. For a complex topologically simple Banach algebra, algebraic pure infiniteness is equivalent to sequential pure infiniteness in the sense of Cortiñas–Montero–Rodríguez together with the existence of one nonzero idempotent. Thus the source’s bounded approximate unit of idempotents can be replaced by a single idempotent, and one implication needs no approximate identity at all. Whether the sequential condition alone forces an idempotent remains open.

1 Source question

The source is K. Bardadyn, B. Kwaśniewski, and A. McKee, *Banach algebras associated to twisted étale groupoids: simplicity and pure infiniteness*, arXiv:2406.05717 [1]. Proposition 6.3(2) compares two notions of pure infiniteness assuming a bounded two-sided approximate unit of idempotents; Remark 6.4 asks for their relationship without that standing assumption.

Proposition 6.3. *Let B be a simple Banach algebra.*

- (1) *If B is unital then the following conditions are equivalent:*
 - (a) *B is a purely infinite Banach algebra;*
 - (b) *B is a purely infinite simple ring in the sense of [AGP02](#), [AAM17](#);*
 - (c) *B is purely infinite in the sense of [Phi13a](#), [DH22](#), that is, $B \not\cong \mathbb{F}$ and for every $y \in B \setminus \{0\}$ there are $x, z \in B$ such that $xyz = 1$.*
- (2) *If B has a bounded (two-sided) approximate unit consisting of idempotents, then the following conditions are equivalent:*
 - (a) *B is a purely infinite Banach algebra;*
 - (b) *B is purely infinite in the sense of [CMR23](#), that is, $B \not\cong \mathbb{F}$ and for every $a, b \in B \setminus \{0\}$ there are sequences $(x_n)_{n=1}^\infty, (z_n)_{n=1}^\infty$ in B such that $x_n b z_n \rightarrow a$;*
 - (c) *for every idempotent $e \in B \setminus \{0\}$, the unital Banach algebra eBe is purely infinite in the sense of the equivalent conditions in [\(1\)](#).*
- (3) *If B is a C^* -algebra, then B is a purely infinite Banach algebra if and only if it is purely infinite in the sense of Cuntz [Cum81](#), that is, every non-zero hereditary C^* -subalgebra of B contains a partial isometry u with $uu^* < u^*u$.*

Figure 1: The standing approximate-unit hypothesis in source Proposition 6.3, PDF page 27.

Remark 6.4. Without the standing assumption in (2) above, we do not know the relationship between pure infiniteness in our sense and that of [CMR23]. When B is a simple unital C^* -algebra, the equivalence between the Cuntz definition and the condition in (1)(b) is well known, see [Bla98 Proposition 6.11.5].

Figure 2: The open comparison in source Remark 6.4, PDF page 28.

The question may be transcribed as follows:

Without the bounded two-sided approximate-unit-of-idempotents assumption, what is the relationship between the algebraic infinite-idempotent notion of pure infiniteness and the sequential sandwich-approximation notion of Cortiñas–Montero–Rodríguez?

2 Definitions

All Banach algebras below are complex. A Banach algebra B is *topologically simple* if its only closed two-sided ideals are 0 and B . Idempotents $e, f \in B$ are equivalent, written $e \sim f$, if $e = uv$ and $f = vu$ for some $u, v \in B$. An idempotent e is *infinite* if it is equivalent to a proper sub-idempotent of itself.

For a topologically simple Banach algebra B , consider:

(API) For every $b \in B \setminus \{0\}$, there are $x, z \in B$ such that xbz is an infinite idempotent.

(SPI) $B \not\cong \mathbb{C}$, and for every $b \in B \setminus \{0\}$ and $a \in B$ there are sequences (x_n) and (z_n) in B such that $x_n b z_n \rightarrow a$.

(API) is Definition 6.1 of the source. (SPI) is the simple pure-infiniteness condition of Cortiñas–Montero–Rodríguez [2]; allowing $a = 0$ does not change it.

3 Proof intuition

There are two complementary mechanisms. Under (API), factor an infinite idempotent e through a chosen nonzero b . The corner eBe is unital purely infinite and hence properly infinite. Its Cuntz families turn every finite sum $\sum c_i e d_i$ into a *single* sandwich Ced . Topological simplicity makes such finite sums dense, giving (SPI).

Conversely, suppose (SPI) and retain only one nonzero idempotent p . The CMR corner theorem makes pBp unital purely infinite, so p itself is infinite. Sequentially factor any nonzero b toward p , compress to pBp , and use openness of its invertible group to correct one sufficiently good approximation to an exact factorization of p through b .

The theorem therefore isolates the full unresolved issue: whether an (SPI) Banach algebra can be idempotent-free.

4 Exact characterization

Theorem 1 (The idempotent gap). *Let B be a complex topologically simple Banach algebra. Then the following are equivalent:*

1. B satisfies (API);

2. B satisfies (SPI) and contains a nonzero idempotent.

In particular, (API) always implies (SPI), without any approximate-identity hypothesis.

We use two established corner facts. First, the source's proof of Proposition 6.3 shows without any approximate-unit hypothesis that if B satisfies (API) and $0 \neq e = e^2 \in B$, then eBe is a unital purely infinite Banach algebra [1, Proposition 6.3(2), implication (a) $\Rightarrow$ (c)]. Second, if B satisfies (SPI), then every nonzero idempotent corner eBe is sequentially purely infinite and is not a division algebra [2, Lemmas 3.1–3.2 and Corollary 3.3]. A unital purely infinite complex Banach algebra is properly infinite [3, Lemma 1.2].

Proof. Assume first that B satisfies (API), and fix $b \neq 0$. Choose $r, s \in B$ such that

$$e = rbs$$

is an infinite idempotent. By the first corner fact, eBe is a unital purely infinite Banach algebra with unit e , hence e is properly infinite. Consequently, for every $m \geq 1$ there are pairwise orthogonal idempotents $e_1, \dots, e_m \leq e$, all equivalent to e . Choose $u_i, v_i \in eBe$ so that

$$u_i v_i = e, \quad v_i u_i = e_i.$$

Orthogonality gives

$$u_i v_j = u_i e_j e_j v_j = 0 \quad (i \neq j), \quad u_i v_i = e. \quad (1)$$

The closure of the algebraic ideal BeB is a nonzero closed two-sided ideal, so topological simplicity gives

$$\overline{BeB} = B. \quad (2)$$

Given $a \in B$, choose finite sums

$$a_n = \sum_{i=1}^{m_n} c_{n,i} e d_{n,i} \longrightarrow a.$$

For a family satisfying (1) with $m = m_n$, set

$$C_n = \sum_{i=1}^{m_n} c_{n,i} u_i, \quad D_n = \sum_{i=1}^{m_n} v_i d_{n,i}.$$

Then (1) yields

$$C_n e D_n = \sum_{i,j} c_{n,i} u_i v_j d_{n,j} = \sum_i c_{n,i} e d_{n,i} = a_n.$$

Since $e = rbs$, we have

$$(C_n r) b (s D_n) = a_n \longrightarrow a.$$

Thus B satisfies (SPI). It also contains the nonzero idempotent e .

Conversely, assume (SPI) and let $0 \neq p = p^2 \in B$. By the second corner fact, pBp is a unital purely infinite Banach algebra; in particular its unit p is an infinite idempotent. Fix $b \neq 0$. By (SPI), choose $x_n, z_n \in B$ such that $x_n b z_n \rightarrow p$. In the unital Banach algebra pBp ,

$$w_n := p x_n b z_n p \longrightarrow p.$$

For some n , $w = w_n$ is invertible in pBp . Therefore

$$(w^{-1} p x_n) b (z_n p) = p.$$

The right-hand side is an infinite idempotent, proving (API). □

Corollary 2. *The unrestricted equivalence of (API) and (SPI) holds if and only if every (SPI) complex Banach algebra contains a nonzero idempotent. Any counterexample to the equivalence must be idempotent-free.*

Proof. The forward assertion is immediate from Theorem 1. Conversely, (API) itself produces a nonzero infinite idempotent, so an idempotent-free (SPI) algebra cannot satisfy (API). $\square$

5 Scope, verification, and novelty

The result is an exact characterization, but it does not decide whether the extra idempotent condition follows automatically from (SPI). Attempts using approximate generalized inverses, unitization and spectral calculus, primitive/radical structure, matrix amplification, and projectionless examples did not remove that last condition. In particular, (SPI) provides approximations $x_n b z_n \rightarrow b$, not the regularity relation $by_n b \rightarrow b$ needed for standard idempotent perturbation arguments.

Verification notes. There is no computational dependency. Review should focus on three points: (i) the source’s corner implication $(\text{API}) \Rightarrow$ unital pure infiniteness of eBe uses no approximate unit; (ii) proper infiniteness gives the relations (1) for every finite size; and (iii) the CMR corner results make pBp unital purely infinite under (SPI). The remaining steps are finite-sum algebra and the Neumann-series openness of the invertible group of pBp .

Novelty check. On 11 August 2026, the four run indexes were searched for arXiv:2406.05717 and the terms “pure infiniteness”, “CMR”, and “idempotent”. Bounded arXiv/web searches used the exact source phrase and combinations of “simple purely infinite Banach algebra”, “nonzero idempotent”, and “approximate unit”. The source, CMR arXiv:2307.05555, and Daws–Horváth arXiv:2104.14989 were checked directly. They contain the component corner and proper-infiniteness facts, but no statement of Theorem 1 was located. Novelty confidence is moderate because the combination is short and may be folklore.

Human-review recommendation. Review as a substantial partial result. It removes essentially all of the source’s approximate-unit hypothesis and identifies the exact residual obstruction, but should not be presented as proving that (API) and (SPI) are unconditionally equivalent.

References

- [1] K. Bardadyn, B. Kwaśniewski, and A. McKee, *Banach algebras associated to twisted étale groupoids: simplicity and pure infiniteness*, arXiv:2406.05717, Proposition 6.3 and Remark 6.4.
- [2] G. Cortiñas, D. Montero, and M. E. Rodríguez, *Simplicity of L^p -graph algebras*, arXiv:2307.05555, Section 3.
- [3] M. Daws and B. Horváth, *A purely infinite Cuntz-like Banach $*$ -algebra with no purely infinite ultrapowers*, J. Math. Anal. Appl. 514 (2022), 126241; arXiv:2104.14989.